Assignment - Confusion Matrix

Results from Random Forest

```
In [20]: print(cm)
[[78  7]
 [ 6 43]]

In [21]: from sklearn.metrics import classification_report
clf_report = classification_report(y_test, y_pred)

In [22]: print(clf_report)
              precision    recall  f1-score   support

           0       0.93      0.92      0.92        85
           1       0.86      0.88      0.87        49

    accuracy                           0.90       134
   macro avg       0.89      0.90      0.90       134
weighted avg       0.90      0.90      0.90       134
```

Class 0 represents Not Purchased, and class 1 represents Purchased. Rows show actual classes; columns show predicted classes.

Actual ↓ / Predicted →	Not Purchased (0)	Purchased (1)	Actual total
Not Purchased (0)	**78**	7	**85**
Purchased (1)	6	**43**	**49**
Predicted total	**84**	**50**	**134**

78: Correctly predicted as not purchased
06: Actual purchasers incorrectly predicted as not purchased
43: Correctly predicted as purchased
07: Actual non-purchasers incorrectly predicted as purchased

Accuracy: Out of all customers, what percentage did the Random Forest classify correctly? 90% - Approximately 90% of the 134 customers were correctly classified as Purchased or Not purchased.

What proportion of all predictions were correct?

Calculation: (78+43) / (78+7+6+43) => 121/134 => 90%

Precision-0: Of the customers the model predicted as not purchased, what percentage actually did not purchase? 93% - Approximately 93% of its Not purchased prediction were correct.

How reliable are the predictions for each class?

Calculation: (78 / (78+6) => 78/84 => 93%

Precision-1: Of the customers the model predicted as purchased, what percentage actually did purchase? 86% - Approximately 86% of its purchased prediction were correct.

How reliable are the predictions for each class?

Calculation: (43 / (43+7) => 43/50 => 86%

Recall-0: Of all customers who actually did not purchase, what percentage did the model correctly identify? 92% the model identified approximately 92% of actual non-purchases and missed approximately 8%

What proportion of actual cases did the model identify?

Calculation: (78 / (78+7) => 78/85 => 92%

Recall-1: Of all customers who actually purchased, what percentage did the model correctly identify? 88% the model identified approximately 88% of actual purchases and missed approximately 12%

What proportion of actual cases did the model identify?

Calculation: (43 / (43+6) => 43/49 => 88%

F1 score 0: How well does the model balance precision and recall for Not Purchased? 92% Its combined precision and recall score for non-purchasers is approximately 92%.

How well are precision and recall balanced?

Calculation: (2*Recall*Precision) / (Recall + Precision) => (2*93%*92%) /(93%+92%) => 92%

F1 score 1: How well does the model balance precision and recall for Purchased? 87% Its combined precision and recall score for purchasers is approximately 87%.

How well are precision and recall balanced?

Calculation: (2*Recall*Precision) / (Recall + Precision) => (2*88%*86%) /(88%+86%) => 87%

Support 0: How many customers in the test data actually did not purchase? 85 -There are 85 actual non-purchasers in y_test.

How many actual cases belong to each class?

Calculation: (78+7) = 85

Support 1: How many customers in the test data actually purchased? 49 -There are 49 actual purchasers in y_test.

How many actual cases belong to each class?

Calculation: (43+6) = 49

(Support counts actual cases, not predicted cases.)

Macro precision: What is the average precision when both classes receive equal important? 89% - Average of the two class precision scores.

Calculation: (93%+86%)/2 = 89%

Macro recall: What is the average recall when both classes receive equal important? 90% - Average of the two class recall scores.

Calculation: (92%+88%)/2 = 90%

Macro F1: What is the average F1 score when both classes receive equal important? 90% - Average of the two class F1 scores.

Calculation: (92%+87%)/2 = 90%

Weighted precision: What is the average precision after accounting for each class's actual count? 90% Class 0 contributes more because it has 85 cases, compared with 49 for class 1.

What is the average after considering actual class counts?

Calculation: (85/134) *(78/84) + (49/134)*(43/50) => (0.634328*(0.928571) + (0.365672)*(0.86) = 90%

Weighted recall: What is the average recall after accounting for each class's actual count? 90% - Recall weighted by the actual class count.

Calculation: (85/134) *(78/85) + (49/134)*(43/49) => (0.634328*(0.917647) + (0.365672)*(0.8776) = 90%

Weighted F1: What is the average F1 score after accounting for each class's actual count? 90% - F1 weighted by the actual class count.

Calculation: (85/134) *92% + (49/134)*87% => (0.634328*(0.923077) + (0.365672)*(0.868687) => 0.903188 = 90%

Results from SVM

```
In [16]: print(clf_report)
```

```
              precision    recall  f1-score   support

           0       0.76      0.96      0.85        85
           1       0.88      0.47      0.61        49

    accuracy                           0.78       134
   macro avg       0.82      0.72      0.73       134
weighted avg       0.81      0.78      0.76       134
```

```
In [17]: print(cm)
```

```
[[82  3]
 [26 23]]
```

```
In [18]: classifier.predict([[40,300,1,]])
```

Class 0 represents Not Purchased, and class 1 represents Purchased. Rows show actual classes; columns show predicted classes.

Actual ↓ / Predicted →	Not Purchased (0)	Purchased (1)	Actual total
Not Purchased (0)	82	3	**85**
Purchased (1)	26	23	**49**
Predicted total	**108**	26	**134**

82: Correctly predicted as not purchased
26: Actual purchasers incorrectly predicted as not purchased
23: Correctly predicted as purchased
03: Actual non-purchasers incorrectly predicted as purchased

Accuracy: Out of all customers, what percentage did the SVM classify correctly? 78% - Approximately 78% of the 134 customers were correctly classified as Purchased or Not purchased.

What proportion of all predictions were correct?

Calculation: (82+23) / (82+3+23+26) => 105/134 => 78%

Precision-0: Of the customers the model predicted as not purchased, what percentage actually did not purchase? 76% - Approximately 76% of its Not purchased prediction were correct.

How reliable are the predictions for each class?

Calculation: (82 / (82+26) => 82/108 => 76%

Precision-1: Of the customers the model predicted as purchased, what percentage actually did purchase? 88% - Approximately 88% of its purchased prediction were correct.

How reliable are the predictions for each class?

Calculation: (23 / (23+3) => 23/26 => 88%

Recall-0: Of all customers who actually did not purchase, what percentage did the model correctly identify? 96% the model identified approximately 96% of actual non-purchases and missed approximately 4%

What proportion of actual cases did the model identify?

Calculation: (82/ (82+3) => 82/85 => 96%

Recall-1: Of all customers who actually purchased, what percentage did the model correctly identify? 47% the model identified approximately 47% of actual purchases and missed approximately 53%

What proportion of actual cases did the model identify?

Calculation: (23 / (23+26) => 23/49 => 47%

F1 score 0: How well does the model balance precision and recall for Not Purchased? 85% Its combined precision and recall score for non-purchasers is approximately 85%.

How well are precision and recall balanced?

Calculation: (2*Recall*Precision) / (Recall + Precision) => (2*96%*76%) /(96%+76%) => 85%

F1 score 1: How well does the model balance precision and recall for Purchased? 61% Its combined precision and recall score for purchasers is approximately 61%.

How well are precision and recall balanced?

Calculation: (2*Recall*Precision) / (Recall + Precision) => (2*47%*88%) /(47%+88%) => 61%

Support 0: How many customers in the test data actually did not purchase? 85 -There are 85 actual non-purchasers in y_test.

How many actual cases belong to each class?

Calculation: (82+3) = 85

Support 1: How many customers in the test data actually purchased? 49 -There are 49 actual purchasers in y_test.

How many actual cases belong to each class?

Calculation: (26+23) = 49

(Support counts actual cases, not predicted cases.)

Macro precision: What is the average precision when both classes receive equal important? 82% - Average of the two class precision scores.

Calculation: (76%+88%)/2 = 82%

Macro recall: What is the average recall when both classes receive equal important? 72% - Average of the two class recall scores.

Calculation: (96%+47%)/2 = 72%

Macro F1: What is the average F1 score when both classes receive equal important? 73% - Average of the two class F1 scores.

Calculation: (85%+61%)/2 = 73%

Weighted precision: What is the average precision after accounting for each class's actual count? 85 Class 0 contributes more because it has 85 cases, compared with 49 for class 1.

What is the average after considering actual class counts?

Calculation: (85/134) *(82/108) + (49/134)*(23/26) => (0.634328*(0.759259) + (0.365672)*(0.884615) =>0.805098 =81%

Weighted recall: What is the average recall after accounting for each class's actual count? 78% - Recall weighted by the actual class count.

Calculation: (85/134) *(82/85) + (49/134)*(23/49) => (0.634328*(0.964706) + (0.365672)*(0.469388) => 0.783582 = 78%

Weighted F1: What is the average F1 score after accounting for each class's actual count? 90% - F1 weighted by the actual class count.

Calculation: (85/134) *85% + (49/134)*61% => (0.634328*(0.849741) + (0.365672)*(0.613333) => 0.763293 = 76%

Results from Decision Tree

```
In [16]: print(clf_report)
```

```
              precision    recall  f1-score   support

           0       0.90      0.89      0.90        85
           1       0.82      0.84      0.83        49

    accuracy                           0.87       134
   macro avg       0.86      0.87      0.86       134
weighted avg       0.87      0.87      0.87       134
```

```
In [17]: print(cm)
```

```
[[76  9]
 [ 8 41]]
```

Class 0 represents Not Purchased, and class 1 represents Purchased. Rows show actual classes; columns show predicted classes.

Actual ↓ / Predicted →	Not Purchased (0)	Purchased (1)	Actual total
Not Purchased (0)	76	9	**85**
Purchased (1)	8	41	**49**
Predicted total	84	50	**134**

76: Correctly predicted as not purchased
08: Actual purchasers incorrectly predicted as not purchased
41: Correctly predicted as purchased
09: Actual non-purchasers incorrectly predicted as purchased

Accuracy: Out of all customers, what percentage did the Decision Tree classify correctly? 87% - Approximately 87% of the 134 customers were correctly classified as Purchased or Not purchased.

What proportion of all predictions were correct?

Calculation: (76+41) / (76+9+41+8) => 117/134 => 87%

Precision-0: Of the customers the model predicted as not purchased, what percentage actually did not purchase? 90% - Approximately 90% of its Not purchased prediction were correct.

How reliable are the predictions for each class?

Calculation: (76 / (76+8) => 76/84 => 90%

Precision-1: Of the customers the model predicted as purchased, what percentage actually did purchase? 82% - Approximately 82% of its purchased prediction were correct.

How reliable are the predictions for each class?

Calculation: (41 / (41+9) =>41/50 => 82%

Recall-0: Of all customers who actually did not purchase, what percentage did the model correctly identify? 89% the model identified approximately 89% of actual non-purchases and missed approximately 11%

What proportion of actual cases did the model identify?

Calculation: (76/ (76+9) => 76/85 => 89%

Recall-1: Of all customers who actually purchased, what percentage did the model correctly identify? 84% the model identified approximately 84% of actual purchases and missed approximately 16%

What proportion of actual cases did the model identify?

Calculation: (41 / (41+8) => 41/49 => 84%

F1 score 0: How well does the model balance precision and recall for Not Purchased? 90% Its combined precision and recall score for non-purchasers is approximately 90%.

How well are precision and recall balanced?

Calculation: (2*Recall*Precision) / (Recall + Precision) => (2*89%*90%) /(89%+90%) => 90%

F1 score 1: How well does the model balance precision and recall for Purchased? 83% Its combined precision and recall score for purchasers is approximately 83%.

How well are precision and recall balanced?

Calculation: (2*Recall*Precision) / (Recall + Precision) => (2*84%*82%) /(84%+82%) => 83%

Support 0: How many customers in the test data actually did not purchase? 85 -There are 85 actual non-purchasers in y_test.

How many actual cases belong to each class?

Calculation: (76+9) = 85

Support 1: How many customers in the test data actually purchased? 49 -There are 49 actual purchasers in y_test.

How many actual cases belong to each class?

Calculation: (41+8) = 49

(Support counts actual cases, not predicted cases.)

Macro precision: What is the average precision when both classes receive equal important? 86% - Average of the two class precision scores.

Calculation: (90%+82%)/2 = 86%

Macro recall: What is the average recall when both classes receive equal important? 87% - Average of the two class recall scores.

Calculation: (89%+84%)/2 = 87%

Macro F1: What is the average F1 score when both classes receive equal important? 86% - Average of the two class F1 scores.

Calculation: (89%+83%)/2 = 86%

Weighted precision: What is the average precision after accounting for each class's actual count? 87% Class 0 contributes more because it has 85 cases, compared with 49 for class 1.

What is the average after considering actual class counts?

Calculation: (85/134) *(76/84) + (49/134)*(41/50) => (0.634328*(0.904762) + (0.365672)*(0.82000) =>0.873767=87%

Weighted recall: What is the average recall after accounting for each class's actual count? 87% - Recall weighted by the actual class count.

Calculation: (85/134) *(76/85) + (49/134)*(41/49) => (0.634328*(0.894118) + (0.365672)*(0.836735) => 0.873134= 87%

Weighted F1: What is the average F1 score after accounting for each class's actual count? 87% - F1 weighted by the actual class count.

Calculation: (85/134) *89% + (49/134)*83% => (0.634328*(0.899408) + (0.365672)*(0.828283) => 0.8734 = 87%

Combined Results

Metric	Random Forest	SVM	Decision Tree
Accuracy	90.30%	78.36%	87.31%
Precision 0	92.86%	75.93%	90.48%
Precision 1	86.00%	88.46%	82.00%
Recall 0	91.76%	96.47%	89.41%
Recall 1	87.76%	46.94%	83.67%
F1 0	0.923077	0.849741	0.899408
F1 1	0.868687	0.613333	0.828283
Macro precision	89.43%	82.19%	86.24%
Macro recall	89.76%	71.70%	86.54%
Macro F1	0.895882	0.731537	0.863846
Weighted precision	90.35%	80.51%	87.38%
Weighted recall	90.30%	78.36%	87.31%
Weighted F1	0.903188	0.763293	0.873400
Support 0	85	85	85
Support 1	49	49	49